SRI MANAKULA VINAYAGAR
ENGINEERING COLLEGE
(AN AUTONOMOUS INSTITUTION)
PUDUCHERRY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Title of the Project:	**FINGERTIP TISSUE DEFORMABILITY BASED NON-INVASIVE ANEMIA DETECTION DEVICE**
Name and Designation of the Supervisor:	**Dr. R. KALAIYARASAN** **ASSOCIATE PROFESSOR**

S. No	Name of the Student	Register Number	Year /Sem/Sec
1	MENAGA S	22UEC113	III/VI/A
2	VASUNDARAA J N	22UEC209	III/VI/A

ABSTRACT

The proposed work presents the design and implementation of a non-invasive anemia detection system based on fingertip tissue deformability and optical sensing techniques. Conventional anemia diagnosis relies on invasive blood sampling and laboratory analysis, which can be time-consuming, costly, and uncomfortable for patients. To overcome these limitations, the system uses a heart pulse sensor to measure variations in light absorption caused by blood flow in the fingertip, along with a load cell to apply controlled pressure. This pressure alters tissue thickness and blood distribution, resulting in measurable changes in the optical signal. An Arduino Uno microcontroller is used to acquire, process, and analyze both baseline and pressure-induced signals using filtering and averaging techniques to reduce noise and improve reliability. Based on calibrated thresholds, the system estimates hemoglobin-related variations and classifies the condition into normal, mild anemia, or severe anemia. The results, including pulse value, pressure level, and anemia status, are displayed on an LCD for real-time user feedback. The proposed system is compact, low-cost, portable, and easy to use, making it suitable for preliminary anemia screening in rural healthcare, medical camps, and home monitoring.

PROJECT DOMAIN

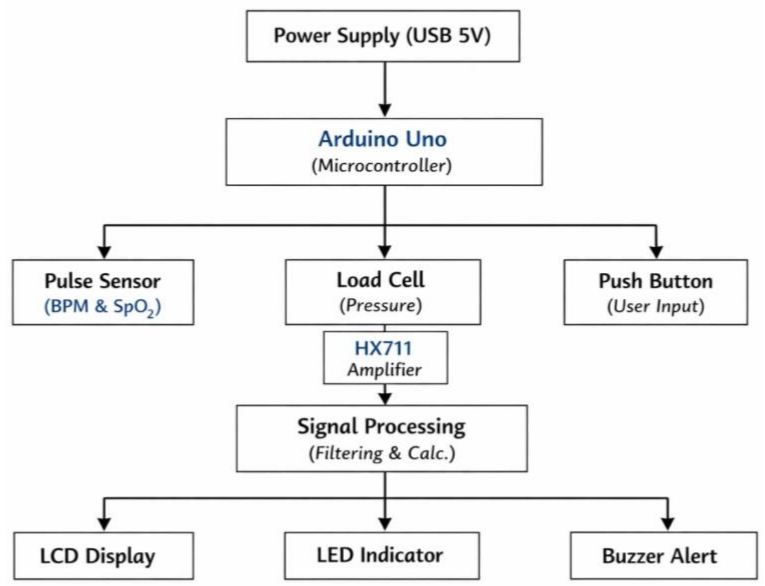
BIOMEDICAL, EMBEDDED SYSTEM

APPLICATION OF THE PROPOSED SYSTEM

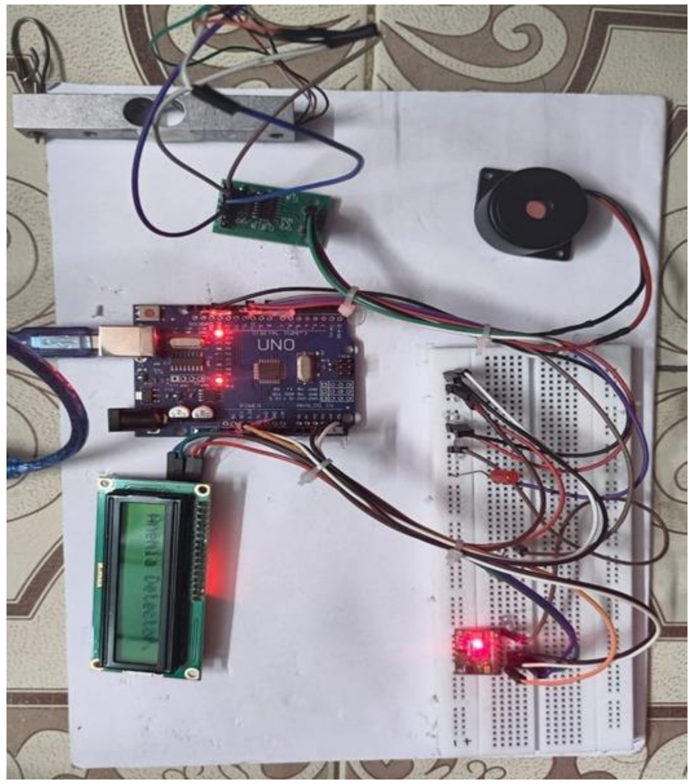
- **Primary Healthcare Screening –** Enables quick and non-invasive anemia detection in rural and low-resource areas.
- **Home Health Monitoring –** Allows individuals to monitor their health condition regularly without visiting laboratories.
- **Medical Camps and Awareness Programs –** Useful for large-scale screening in public health initiatives.
- **Educational and Research Use –** Helps students understand biomedical sensing, signal processing, and embedded system design.

PHOTOGRAPHS OF THE PROPOSED SYSTEM

BLOCK DIAGRAM



PROPOSED SYSTEM SOFTWARE /HARDWARE IMAGE



BATCH : 2022-2026